



**BUSINESS
PROFESSIONALS**
of **AMERICA**
Giving Purpose to Potential



SPREADSHEET APPLICATIONS (230)

STATE - 2020

Production Portion:

Job 1: Create Spreadsheet with Formulas _____ (160 points)

Job 2: Format Spreadsheet _____ (70 points)

Job 3: Create Chart _____ (70 points)

TOTAL POINTS _____ ***(300 points)***

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

No more than ten (10) minutes orientation
No more than ninety (90) minutes testing time
No more than ten (10) minutes wrap-up

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Workplace Skills Assessment Program competition.

GENERAL INSTRUCTIONS

1. Make certain this test booklet contains Jobs 1-3.
2. Key all jobs according to the instructions given.
3. Correct any obvious keyboarding errors and incorporate any editing notations.
4. Correct any errors in formatting. Use formatting shown in the *Style & Reference Manual*.
5. For any problem where you would normally key your reference initials, key your contestant number. Your name or initials should *not* appear on any work you submit.
6. Key your contestant number and job number as the footer in the lower left-hand corner of all work submitted unless otherwise specified.

Example:
99-9999-9999 Job 1

7. If you finish before the end of the testing time, notify the proctor. Time may be a factor in determining a winner when there is a tie score.
8. When turning in your materials, place your scoring sheet on top of your jobs. The jobs should be arranged in numerical order. Turn in all partial jobs completed
9. Center all worksheets vertically and horizontally to print on one page.
10. Print all worksheets in Landscape format.
11. Print each chart on a separate sheet.

Scenario

Your organization is presenting to a group of eighth graders about careers. Your part of the presentation is creating the spreadsheet comparing wages between different education levels. Using the information you gathered from Bureau of Labor Statistics create the following spreadsheet.

GRADER: points are all or none unless otherwise noted

Job 1 (Worksheet with formulas)	Points Possible	Points Earned
Contestant # and Job # in left section of footer	5	
12 pt. Times New Roman font used for data	5	
Inserted Row before high school graduate and entered: high school dropout; 13.83;	5	
Inserted Row before college graduate and entered: some college, no degree; 20.05;	5	
Update cells (5 points each): - A14 and A19: High School dropout and graduate - A15 and A20: High School graduate and 2 year graduate - A16 and A21: 2 year college graduate and a 4 year graduate	15	
Used PAY_PER_HOUR and HOURS in formula in cell C4 that calculates gross pay; Formula filled down for all other education levels	10	
Used TAX_RATE in formula used in cell D4 to calculate the net pay after taxes; Fill formula down for all other education levels	10	
E4 formula for gross pay for 2 weeks; fill the formula down for the other education levels	10	
F4 formula for gross pay for a month; fill the formula down for the other education levels	10	
Cell B14 create a IF function calculating the difference between high school dropout and high school graduate	10	
Cell B15 create a IF function calculating the difference between high school graduate and 2 year graduate	10	
Cell B16 create a IF function calculating the difference between 2 year graduate and 4 year graduate	10	
Cell B19 create a formula calculating the difference in yearly salary between high school dropout and high school graduate.	10	
Cell B20 create a formula calculating the difference in yearly salary between high school graduate and 2 year graduate	10	
Cell B21 create a formula calculating the difference in yearly salary between 2 year graduate and four year graduate	10	
Column widths adjusted to see all data	5	
Print out is landscape, centered vertically and horizontally, on one page, with gridlines and headings	10	
Print out of formulas in landscape view with the worksheet centered on ONE page with gridlines and headings	10	
Subtotal	160	

Job 2 (Formatted worksheet)	Points Possible	Points Earned
Contestant # and Job # in left section of footer	5	
Printed in landscape, centered vertically and horizontally on one sheet	10	
B4:F11; B13:B16; and B19:B21 currency with 2 decimal places (5 points each range)	15	
A1:F1: merge and center across the spreadsheet, bold, font size 18	5	
A3:F3: center and bold	5	
A13:B13 and A18:B18: merge and center and bold	5	
A14:A16 and A19:A21: right align	5	
Row 3 has Text Wrap applied	5	
Column width A is 50pts and Columns B:F are 12pts wide	5	
Shading applied to A3:F3, B13, and B18	5	
Thick outside border around cells A1:F11, A13:B16, and A18:B21	5	
Subtotal	70	

Job 3 (Clustered Column Chart)	Points Possible	Points Earned
Contestant # and Job # in left section of footer	5	
Correct Clustered Column Chart created on its own sheet	15	
Printed on full sheet, landscape in Black & White	10	
Correct title entered	10	
Format chart to style 2	5	
Format chart to layout 1	5	
Legend Removed	10	
Data Callout Data Label style	10	
Subtotal	70	
TOTAL POINTS	300	

SPREADSHEET APPLICATIONS - State 2020 - KEY
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JOB 1: Create Spreadsheet with Formulas

	A	B	C	D	E	F
1	2018 EARNING POWER BASED ON EDUCATIONAL LEVEL					
2						
3	EDUCATION LEVEL	PAY PER HOUR	GROSS PAY	NET PAY AFTER TAXES	GROSS PAY FOR 2 WEEKS	MONTHLY GROSS PAY
4	High School Dropout	13.83	553.2	453.624	907.248	1814.496
5	High School Graduate	18.25	730	598.6	1197.2	2394.4
6	Some College, No Degree	20.05	802	657.64	1315.28	2630.56
7	2-Year College Graduate (Associates Degree)	21.55	862	706.84	1413.68	2827.36
8	4-Year College Graduate (Bachelor's Degree)	29.95	1198	982.36	1964.72	3929.44
9	Master's Degree	35.85	1434	1175.88	2351.76	4703.52
10	Professional Degree	47.1	1884	1544.88	3089.76	6179.52
11	Doctoral Degree	45.63	1825.2	1496.664	2993.328	5986.656
12						
13	Difference in net pay for the week between:					
14	High School Dropout and graduate	144.976				
15	High School Graduate and 2- year College Graduate	108.24				
16	2-Year College Graduate and 4-Year College Graduate	275.52				
17						
18	Differences in yearly net pay between:					
19	High School Dropout and graduate	3769.376				
20	High School Graduate and 2- year College Graduate	2814.24				
21	2-Year College Graduate and 4-Year College Graduate	7163.52				

Contestant#
Job1

	B	C	D	E	F
EDUCATIONAL LEVEL					
	PAY PER HOUR	GROSS PAY	NET PAY AFTER TAXES	GROSS PAY FOR 2 WEEKS	MONTHLY GROSS PAY
	13.83	=PAY_PER_HOUR*HOURS	=C4-(C4*TAX_RATE)	=D4*2	=D4*4
	18.25	=PAY_PER_HOUR*HOURS	=C5-(C5*TAX_RATE)	=D5*2	=D5*4
	20.05	=PAY_PER_HOUR*HOURS	=C6-(C6*TAX_RATE)	=D6*2	=D6*4
gree)	21.55	=PAY_PER_HOUR*HOURS	=C7-(C7*TAX_RATE)	=D7*2	=D7*4
gree)	29.95	=PAY_PER_HOUR*HOURS	=C8-(C8*TAX_RATE)	=D8*2	=D8*4
	35.85	=PAY_PER_HOUR*HOURS	=C9-(C9*TAX_RATE)	=D9*2	=D9*4
	47.1	=PAY_PER_HOUR*HOURS	=C10-(C10*TAX_RATE)	=D10*2	=D10*4
	45.63	=PAY_PER_HOUR*HOURS	=C11-(C11*TAX_RATE)	=D11*2	=D11*4
reen:					
	=IF(D4>D5,D4-D5,D5-D4)				
ge Graduate	=IF(D5>D7,D5-D7,D7-D5)				
lege Graduate	=IF(D7>D8,D7-D8,D8-D7)				
	=B14*26				
ge Graduate	=B15*26				
lege Graduate	=B16*26				

JOB 2: Format the Spreadsheet you created in Job 1 using the following instructions:

2018 EARNING POWER BASED ON EDUCATIONAL LEVEL

EDUCATION LEVEL	PAY PER HOUR	GROSS PAY	NET PAY AFTER TAXES	GROSS PAY FOR 2 WEEKS	MONTHLY GROSS PAY
High School Dropout	\$13.83	\$553.20	\$453.62	\$907.25	\$1,814.50
High School Graduate	\$18.25	\$730.00	\$598.60	\$1,197.20	\$2,394.40
Some College, No Degree	\$20.05	\$802.00	\$657.64	\$1,315.28	\$2,630.56
2-Year College Graduate (Associates Degree)	\$21.55	\$862.00	\$706.84	\$1,413.68	\$2,827.36
4-Year College Graduate (Bachelor's Degree)	\$29.95	\$1,198.00	\$982.36	\$1,964.72	\$3,929.44
Master's Degree	\$35.85	\$1,434.00	\$1,175.88	\$2,351.76	\$4,703.52
Professional Degree	\$47.10	\$1,884.00	\$1,544.88	\$3,089.76	\$6,179.52
Doctoral Degree	\$45.63	\$1,825.20	\$1,496.66	\$2,993.33	\$5,986.66

Difference in net pay for the week between:

High School Dropout and graduate	\$144.98
High School Graduate and 2- year College Graduate	\$108.24
2-Year College Graduate and 4-Year College Graduate	\$275.52

Differences in yearly net pay between:

High School Dropout and graduate	\$3,769.38
High School Graduate and 2- year College Graduate	\$2,814.24
2-Year College Graduate and 4-Year College Graduate	\$7,163.52

Contestant#
Job2

JOB 3: Create a Clustered Column Chart.

